QPCode **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION OCTOBER 2026

FIFTH SEMESTER

Discipline Specific Course(DSC)-MG5DSCMBG300-AGRICULTURAL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1.5 hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Notes Questions

Answer any **10** questions

Each question carries **2** marks

1. What type of microbial interaction benefits one species while the other is unaffected? [K] / [CO1]
2. What is the photosynthetic partner of a lichen called? [K] / [CO1]
3. Explain amensalism. [U] / [CO1]
4. What type of mycorrhiza forms intracellular structures such as arbuscules inside root cells? [K] / [CO1]
5. List any two importance of phyllosphere microflora. [An] / [CO2]
6. Identify any two siderophore producing bacteria. [A] / [CO2]
7. Point out any two importance of carbon cycle. [An] / [CO2]
8. Give any two examples of endophytic bacteria. [U] / [CO2]
9. Identify the pathogen responsible for crown gall disease. [A] / [CO3]

10. Define IPM. [K] / [CO3]

11. What are elicitors? [K] / [CO3]

12. Give one example for a symbiotic nitrogen fixing biofertilizer. [U] / [CO3]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Explain how commensalism benefits one microorganism with out affecting the other. [U] / [CO1]

14. Describe how ectomycorrhizae improve nutrient uptake by plants. [K] / [CO1]

15. Discuss the microbiology of silage. [C] / [CO2]

16. Analyse the role of microorganisms in the oxidation and reduction steps of sulphur cycle. [An] / [CO2]

17. Explain the symptoms and mode of transmission of TMV. [A] / [CO3]

18. Explain the mechanism of action of Bt toxin against insect pest. [A] [CO3]

[5x4 = 20]

Part C

Essay Question

Answer any **1** questions

Each question carries **10** marks

19. Explain the major groups of microorganisms present in soil and their importance in soil fertility. [U]/ [CO1]

20. Construct a detailed account of Nitrogen cycle by integrating the processes of nitrogen fixation, ammonification, nitrification and denitrification. [C]/ [CO2]

[10x1 = 10]